

The tadpole reabsorption lemma: normal-ordered cubic vertices in the matched bookkeeping eliminate the tadpole channel identically

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1. Purpose and scope

Unit U4 identified the tadpole channel ($9M^2\hat{G}$ contraction of $\langle V_3^2 \rangle_c$) as LOAD-BEARING for any SC-SCOPE lift: its conservative bound exceeds the endpoint composed margin ($\times 0.53$). This note shows the channel is eliminated IDENTICALLY by the matched bookkeeping’s own structure. Scope: the matched bookkeeping of the B5 chain (per-pattern self-consistent Hartree dressing + per-pattern condensate optimization, as executed throughout Math426–Math437 and the B5 chain).

2. The lemma

Lemma 1 (Tadpole reabsorption – sketch). *In the matched bookkeeping, the third-cumulant correction of any competitor pattern is*

$$\delta F_3[\mathcal{P}] = -\frac{1}{2} \langle :V_3:^2 \rangle_c = -\frac{1}{2} \sum_t |\hat{g}_3(t)|^2 6 \widehat{G^3}(t),$$

i.e. the SUNSET channel only: the tadpole contraction $9M^2\hat{G}(t)$ is absent identically.

Proof sketch. (i) Matched bookkeeping evaluates each pattern at its self-consistent Hartree optimum: the dressed propagator G solves the per-pattern gap equation, and the condensate amplitude solves the per-pattern stationarity equation, which contains the Hartree-dressed quartic source acting on the condensate — the $3u_{\text{eff}}MA$ line of the production engine (Math437 §1: “identical to the $0.25UN_4 + 2.5VN_4M$ line”; the dressed coefficient $u_{\text{eff}} = u + 10vM$ is Lemma 1’s object). (ii) Expanding the interaction about this optimum, every Gaussian self-contraction of a vertex leg either renormalizes the propagator (resummed into G by the gap equation) or sources the condensate (absorbed by the stationarity equation). Formally: the perturbation must be written in terms of vertices NORMAL-ORDERED with respect to the dressed Gaussian, $V_3 = \int g_3 : \delta\varphi^3$, because the non-normal-ordered pieces are exactly what the self-consistent dressing already counted. (iii) Wick’s theorem for normal-ordered vertices forbids self-contractions: $\langle : \delta\varphi^3(x) : \delta\varphi^3(y) : \rangle = 3! G(x-y)^3 = 6G^3$. The $9M^2G$ term — which arises precisely from one self-contraction at each vertex — does not occur. Equivalently: the would-be tadpole linear source $3Mg_3 = 3Mu_{\text{eff}}F + O(F^3)$ coincides term-by-term with the stationarity-equation source, and re-including it at third order double-counts the resummation. (iv) The $O(F^3)$ part of g_3 (weight $O(I^{3/2})$) generates a correction to the stationarity point of relative size $O(I)$; its effect on δF_3 is $O(I^4)$, two orders below the margin scale ($< 10^{-6}$ at all anchors by the U4 sextic-fiber arithmetic). $\square$

3. Consequence for the U4 assessment

The U4 table’s tadpole rows ($\times 4.3/ \times 1.5/ \times 0.53$, “if uncanceled”) are STRUCK: the channel does not exist in the matched bookkeeping. The surviving third-cumulant threat is the sunset alone (conservative ratios $\times 7.7/ \times 2.8/ \times 0.97$). The SC-SCOPE lift’s endpoint problem therefore reduces to ONE refinement axis: the per-transfer sunset-kernel decay + cubic-transfer counting (U4 inputs (a)+(b)); input (c) is supplied by this lemma (modulo the registered residuals below).

4. Registered residuals (honest)

- **R-U6-1 (formal alignment writeup)**: the normal-ordering argument is textbook for self-consistent Gaussian expansions; what requires the formal page is its alignment with the chain’s EXACT matched-bookkeeping definitions (which contractions the per-pattern gap + stationarity equations resum, written against the Math424-AddA production decomposition). T3 until written.
- **R-U6-2 (machine cross-check)**: assert that the stationarity-equation source coefficient equals $3u_{\text{eff}}M$ term-by-term against the certified Math436/Math437 closed forms (e.g. reproduce the HEX argmin $\hat{r} = 0.309373$ chain) — one small script, registered (not executable this session; sandbox shell unavailable).
- **Out-of-scope**: nothing here touches the sunset channel, whose endpoint refinement remains the open lift item.

5. Numerical verification

No new machine numbers. The lemma is structural; the single arithmetic statement (the $O(F^3)$ remainder is $O(I^4) < 10^{-6}$) follows from the U4 fiber arithmetic ($\langle F^6 \rangle = O(I^3)$ scale, $116.6 \cdot 2.5 I^3 \cdot \text{kernel} < 10^{-6}$ at $I \leq 2 \times 10^{-3}$) on certified constants.

Quantitative sanity check (mandatory): consistency — the struck tadpole bound at the endpoint (5.0×10^{-3}) was LARGER than the sunset bound (2.75×10^{-3}); had the cancellation gone the other way (sunset struck, tadpole surviving) the lift would be dead at the endpoint — the direction matters and is fixed by the contraction combinatorics ($6G^3$ = full cross-pairings, no self-contraction; $9M^2G$ = the self-contraction pattern), not by magnitude tuning; limit case — for $\mathcal{R}_H$ ($F = 0$) both channels vanish and the lemma is trivially consistent; sign-direction — removing a negative-definite channel can only RAISE competitor free energies, i.e. the lemma moves the lift in the selection-favorable direction, and it does so by construction (no double-resummation), not by optimism.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “Normal-ordering is a choice of bookkeeping; choosing it BECAUSE it kills the dangerous channel is motivated reasoning.” DISMISSED: the choice was made by the chain long before U4 — matched bookkeeping with per-pattern self-consistent dressing IS the executed production scheme (Math426–Math437); the lemma merely states its third-order consequence. The double-counting argument is independent of preference: including both the resummed Hartree source AND the raw tadpole contraction counts the same Wick pairing twice, which is an error in ANY scheme.

- (β) “Competitor patterns in the class-wide statement are quantified over, not individually re-optimized — maybe some are evaluated OFF their stationarity point, where the linear term does not vanish.” VALID-with-mitigation: the chain’s comparison functional evaluates each pattern at its own layer optimum (the races optimize t and m per pattern; Prop A quantifies over ALL trials (t, m) , dominating every off-optimum evaluation from below). For a pattern evaluated off-optimum, the residual linear term generates a SHIFT whose energy gain is bounded by the second-order expansion of the layer functional along the shift direction — an in-layer move that the Prop-A quantification over (t, m) trials already covers for amplitude directions, and which for profile directions is an amended- class move covered by the class quantification. The formal writeup (R-U6-1) will carry this case explicitly.
- (γ) “The $O(F^3)$ remainder estimate assumes generic fibers; resonant triples (sum landing back on a shell mode) have $O(I^{3/2})$ amplitude interfering with the $O(I^{1/2})$ shell source — a cross term of order $O(I^2)$, not $O(I^4)$.” VALID-with-mitigation: the cross term shifts the stationarity SOURCE at relative order $O(I)$ (the F^3 -to- F ratio), so the optimum moves by $O(I)$ and the energy by $O(I^2) \times \text{second-derivative}$ — which is the same order as the second-cumulant terms ALREADY counted in the matched functional (the resonant-triple bookkeeping is exactly the N_4 resonance accounting of the production engine, certified in the Math437 gallery). The genuinely third-order remainder is $O(I^4)$ as stated; the $O(I^2)$ resonant piece is not new. The R-U6-1 writeup will make this two-line accounting explicit.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / tadpole-reabsorption-lemma (unit U6)
Precise statement:	LEMMA (T3 sketch): in the matched bookkeeping the cubic vertex is normal-ordered w.r.t. the dressed Gaussian; the tadpole channel (9 M ² G) of the third-cumulant correction is ABSENT IDENTICALLY (its linear source 3 M u_eff F is the stationarity-equation source already resummed). U4’s tadpole rows are struck; the surviving third-cumulant threat is the sunset alone (x7.7 / x2.8 / x0.97 conservative).
Scope:	Matched bookkeeping of the B5 chain (per-pattern self-consistent Hartree + condensate optimization); all anchors.
Dependencies:	U4 assessment (channel identification + arithmetic); Math437 Lemma 1 / production decomposition (u_eff dressing, stationarity line); Math436 argmin chain (cross-check target).
Evidence grade:	ANALYTIC SKETCH (T3); the $O(I^4)$ remainder line is closed-form on certified constants.
Reproduction command:	(registered follow-up R-U6-2) script asserting the 3 u_eff M coefficient against the Math436/Math437 closed forms.

Expected output: coefficient identity to 1e-10; HEX argmin chain reproduced; JSON artefact under claims/B5-BEYOND-LAYER-BOUND/runs/.

Falsification gate: A matched-bookkeeping derivation exhibiting a NON-cancelling tadpole remainder at $O(I^2)$ or larger beyond the resonant accounting (would demote the lemma and restore U4's pessimistic branch); R-U6-2 coefficient mismatch.

Tier before / after: No tier action. The lemma enters at T3; B5 stays T5-CANDIDATE.

No-overclaim statement: The lemma is a SKETCH pending the R-U6-1 alignment writeup and the R-U6-2 machine cross-check. It does NOT lift SC-SCOPE (the sunset endpoint refinement remains open) and does not change any margin, tier, or gate. PDF build + linter PENDING operator-side.

Next required action: R-U6-1 formal writeup; R-U6-2 script; then the sunset endpoint refinement (per-transfer kernel + transfer counting) as the single remaining SC-SCOPE lift item. Mainline continues (U7).